MINI PROJECT: SMART LED CONTROL SYSTEM

Internship: IoT with AI
Week: 1
Project Title: Smart LED Control System
Platform: Arduino Uno and Tinkercad Circuits

1. Introduction

The Smart LED Control System is a basic Arduino-based project that demonstrates how electronic components can be controlled using a microcontroller. The system uses a push button as an input device and an LED as an output device.

When the push button is pressed, the Arduino detects the input and switches ON the LED. When the button is released, the LED switches OFF.

The project helps understand digital input, digital output, circuit connections and Arduino programming.

2. Objectives

- To understand the working of Arduino Uno.
- To interface a push button with a microcontroller.
- To control an LED using a digital output pin.
- To understand input and output operations.
- To design and test a basic electronic circuit using Tinkercad.

3. Components Required

1. Arduino Uno R3.
2. Breadboard.
3. LED.
4. Push button.
5. 220Ω resistor.
6. Jumper wires.

4. Circuit Connections

Push Button:

- One terminal is connected to Arduino digital pin 2.
- The other terminal is connected to GND.
- The Arduino uses the INPUT_PULLUP configuration.

LED:

- LED anode is connected to digital pin 13 through a 220Ω resistor.
- LED cathode is connected to GND.

5. Working Principle

The Arduino continuously monitors the push button connected to digital pin 2.

The INPUT_PULLUP configuration keeps the input HIGH when the button is released. When the button is pressed, the input becomes LOW because the button connects the pin to GND.

The Arduino checks this input condition and controls the LED accordingly.

- Button pressed: LED ON.
- Button released: LED OFF.

6. Arduino Program

```
int buttonPin = 2;
int ledPin = 13;

void setup() {
 pinMode(buttonPin, INPUT_PULLUP);
 pinMode(ledPin, OUTPUT);
}

void loop() {
 int buttonState = digitalRead(buttonPin);

 if (buttonState == LOW) {
  digitalWrite(ledPin, HIGH);
 } else {
  digitalWrite(ledPin, LOW);
 }
}
```

7. Testing and Results

The circuit was designed and tested using Tinkercad Circuits.

During simulation:

- When the push button was pressed, the LED turned ON.
- When the push button was released, the LED turned OFF.

The simulation demonstrated successful communication between the input device and the output device.

8. Applications

- Basic electronic control systems.
- Push-button operated lighting systems.
- Smart home lighting concepts.
- Microcontroller-based automation.
- Learning digital input and output operations.

9. Future Enhancements

- Control LEDs using a mobile application.
- Add Wi-Fi connectivity using ESP32.
- Control multiple LEDs independently.
- Add automatic lighting using a light sensor.
- Integrate the system into a smart home application.

10. Conclusion

The Smart LED Control System provided practical knowledge of Arduino programming, digital input and output, and electronic circuit design. The project demonstrated how a push button can be used to control an LED through a microcontroller. Tinkercad simulation made it possible to design and test the circuit without physical hardware.